

Mirpur University of Science and Technology (MUST)

Mirpur, Azad Jammu & Kashmir
Department of Software Engineering
Instructor: Engr. Asim Javaid

Software Quality Engineering | Lab Manual

LAB SESSION 2 — Student Instructions

ISO 25010 Quality Attribute Analysis | Pair-Based Lab Activity

PAIR WORK — How to Split the Work

Student A	Student B
Evaluates App 1 individually (Experiment B — all 8 characteristics)	Evaluates App 2 individually (Experiment C — all 8 characteristics)
Both students together: Complete App 1 (All 8), App 2 (All 8), Comparative, Recommendations, Python script, and the written report.	

STEP 1 Setup — Both Students | ⌚ 5 minutes

Open the Excel template: [SQE_Lab2_Excel_Templates.xlsx](#)

Go to the  Session Setup sheet and fill in:

- Your names, student IDs, and section/group
- Choose your application pair (A–F) and record both app names
- Fill in App Version Metadata for both apps (found in App Store / Play Store)
- Tick off the Pre-Lab Checklist

STEP 2 Individual App Evaluation | ⌚ 50 minutes | Use your own phone/device

► STUDENT A — Fill in: Structured — App 1

B.2 Characteristic 1 — Functional Suitability (15%)

B.3 Characteristic 2 — Performance Efficiency (12%)

B.4 Characteristic 3 — Compatibility (10%)

B.5 Characteristic 4 — Usability (15%)

B.6 Characteristic 5 — Reliability (15%)

B.7 Characteristic 6 — Security (15%)

B.8 Characteristic 7 — Maintainability (8%)

B.9 Characteristic 8 — Portability (10%)

→ **Record your Rating (1–5) AND Justification for every characteristic**

→ **Use a stopwatch for Performance timings — actual seconds only, no estimates!**

► **STUDENT B — Fill in: 📊 Structured — App 2**

C.3 Characteristic 1 — Functional Suitability (15%)

C.4 Characteristic 2 — Performance Efficiency (12%)

C.5 Characteristic 3 — Compatibility (10%)

C.6 Characteristic 4 — Usability (15%)

C.7 Characteristic 5 — Reliability (15%)

C.8 Characteristic 6 — Security (15%)

C.9 Characteristic 7 — Maintainability (8%)

C.10 Characteristic 8 — Portability (10%)

→ **Same rules: Rating + Justification for every characteristic**

→ **Use the same device type and network conditions as Student A!**

**STEP
3**

Summary Worksheets — Both Students Together | ⌚ 20 minutes

Excel Sheet to Complete	What to Fill In
App 1 (All 8) sheet	Transfer Student A's 8 ratings and observations. Verify the Weighted Total Score calculates correctly. Run the Self-Check before moving on.
App 2 (All 8) sheet	Transfer Student B's 8 ratings and observations. Verify the Weighted Total Score. Run the Self-Check before moving on.
📊 Comparative sheet	Enter both app names. The ratings table auto-calculates gaps and the Better App column. Copy the APP1_RATINGS and APP2_RATINGS arrays shown — you will need them for Python. Use the Writing Scaffold to draft your analysis paragraphs.
💡 Recommendations sheet	Write 2 recommendations for App 1 (Student A leads) and 2 for App 2 (Student B leads). Follow the formula exactly: "[App] should [specific action] to improve [characteristic] from [X] to [Y]. Currently, [deficiency observed]. This change would [user benefit]."

**STEP
4**

Python Script & Radar Chart — GitHub week2/ folder | ⌚ 25 minutes

In your GitHub repository, create or navigate to the [week2/](#) folder. Then follow these steps:

#	Action	Detail
1.	Create	week2/quality_analysis.py — copy the starter script from Lab Manual Section D.2
2.	Edit	Replace the placeholder ratings with your actual values from the 📊 Comparative sheet → Python Array String section
3.	Run	cd week2 then python quality_analysis.py
4.	Verify	Console output must match your Excel Weighted Total Scores
5.	Check	The radar chart PNG is saved automatically in week2/
6.	Commit	Use the exact commit message shown in the code block below

Exact commit message (copy this):

```
git add week2/quality_analysis.py
git add week2/quality_radar_*.png
git commit -m "Week 2: ISO 25010 analysis script and radar chart for Pair
[Letter]"
git push origin main
```

☒ **week2/ folder must contain:**

- quality_analysis.py — with YOUR actual ratings, not the example values
- quality_radar_[App1]_[App2].png — clearly labelled chart with both app names and scores

STEP 5 Written Report — Both Students Together

Refer to the  Report Outline sheet in Excel OR Section 10 of the Lab Manual. **Write a 7-section PDF report. Minimum 600 words for Sections 6 & 7 combined.**

Sec	Title	Required Content
Section 1	Cover Information	App names, version numbers, platform, OS version, evaluator names, student IDs, pair letter, date of evaluation.
Section 2	App Overviews	One paragraph per app: primary purpose, target users, main features, app category. Note any relevant similarities or differences in purpose.
Section 3	Evaluation Methodology	Rating scale used (1–5 Likert). How each characteristic was tested. Device & OS used. Duration per app. Tools used (stopwatch, settings screenshots, etc.).
Section 4	Completed Checklists	Insert your two completed Excel tables (App 1 All 8 and App 2 All 8) — all 8 characteristics with ratings, weights, and detailed observations. One table per app.
Section 5	Quality Profile Chart	Insert your Python-generated radar chart PNG. Must clearly show both apps, all 8 axes labelled, weighted scores in the legend.
Section 6	Comparative Analysis (min. 3 paragraphs)	Para 1: Overall weighted scores + margin. Para 2: Top 3 quality gaps — explain what they mean for users in practice. Para 3: At least 1 trade-off (e.g. Security vs. Usability). Para 4: Which app you recommend for which use case and why.
Section 7	Recommendations	4 total (2 per app). Follow the formula from  Recommendations sheet. Each must name a specific feature/mechanism, current rating, target rating, and user benefit.

 **Save as PDF, named exactly: SQE_Lab2_[YourName]_[AppPair].pdf**

Example: SQE_Lab2_Ahmad_Ali_PairA.pdf

STEP 6 Submission — Before 02:55 (end of session)

Push to GitHub

Use the git commands from Step 4
Ensure both files are in week2/ folder
Confirm the push was successful

Submit to CMS/LMS

Upload: SQE_Lab2_[YourName]_[AppPair].pdf
Paste your GitHub repo URL in submission notes

Deadline: before end of session (02:55)

COMMON MISTAKES — Avoid These!

Mistake

Fix

X	Rating without evidence — 'it feels secure' is not evidence.	For every rating ask: What did I specifically observe or measure?
X	Estimates for performance — guessing times.	Use a stopwatch. Record actual seconds for every operation.
X	All 4s and 5s — no real app is perfect across all 8 characteristics.	For every 4 or 5, spend 2 extra minutes actively looking for deficiencies.
X	Confusing characteristics — marking an app down for Reliability when it is slow.	Slow = Performance. Crashes = Reliability. Re-read ISO definitions before rating.
X	Vague recommendations — 'App X should improve security.'	Name a specific feature + mechanism + target rating + user benefit.
X	Different test conditions — App 1 on fast WiFi, App 2 on 4G.	Both apps: same device, same OS, same day, same network conditions.
X	Brand bias — rating a well-known app higher due to reputation.	Rate what you observe, not what you expect. Cover the app name and re-check.

Good luck! 🍀 — Engr. Asim Javaid | Software Quality Engineering | Lab Session 2